

Project Topic: Nanaimo Bus App

Many people in Nanaimo, especially students, use the bus system as their main form of transportation. But currently there are no well-designed apps to help them find bus routes or receive alerts/updates about the routes they use every day. I have personally found Google Maps and the BC Transit App to be very unhelpful on numerous occasions. This project will fix that problem by using MongoDB to redesign how bus routes are stored and updated, which will make it much easier for users to find the best bus routes and never worry about being late or missing their bus.

The database will need to handle the following kinds of data:

- **User accounts** - usernames, emails, passwords, etc.
- **Saved locations** - addresses, coordinates, labels, etc.
- **Bus routes** - route numbers, departure and arrival locations, etc.
- **Bus stops** - coordinates, names of stops, whether it has a shelter, etc.
- **Saved trips** - origins, destinations, departure times, arrival times, which routes to take, alert settings, etc.

Reason for Using MongoDB

MongoDB is the best choice for this system because it allows for geospatial queries and graph lookups. The graph lookup feature is especially important because a bus route is essentially a directed graph with bus stops as the vertices and the time between stops as the weighted edges. A SQL database would struggle to replicate a graph, and it would be a lot more cumbersome to represent bus routes in a table.

Backup Project Topic: Movie & TV Show Review Site

In case building purely a database for a bus app proves too complicated or too difficult for the purposes of this course, I've chosen a Movie and TV Show Review Site as my backup topic. Review sites are fairly easy to build (I've built a few before), and I'm already familiar with an API I could use to source data for testing.